AI Crypto Hedge Fund - Final Notebook

Execution mode: **FULL FINAL NOTEBOOK**.

This notebook is the single end-to-end reviewer narrative. It imports repository package code and reads committed validation/final-test artifacts. It does not place orders, download live data, call an external LLM or rerun/tune final-test strategies.

Notebook execution context

Metric	Value
Final-test lock	c33b5eb396f6...43af5e
Final-test exposure	EXPOSED
Final-test dir	artifacts/final_test/c33b5eb396f6
Notebook mode	FULL FINAL NOTEBOOK

1. Executive summary

This is a reproducible historical research MVP for an AI-assisted crypto fund workflow. It is long-only, unlevered, daily spot, USDT-cash based, educational and does not enable live order submission. Net results after fees and slippage are the primary evidence.

Frozen final selected results

Level	Selected	Net	Bench	Sharpe	MDD	Cost
level_1	SMA baseline	-7.4%	-5.4%	-0.171	-18.5%	$8,906
level_2	Ensemble	-0.6%	-5.4%	-0.522	-1.4%	$1,600
level_3	CVaR	-18.0%	-25.4%	-0.023	-45.2%	$1,493
level_4	Calendar	-4.1%	-9.3%	-0.880	-9.1%	$3,584
level_5	Large universe	-28.0%	-45.2%	-0.218	-42.2%	$110,939

Validation versus final

Level	Split	Selected	Net	Bench	Sharpe	MDD
Level 1	Validation	SMA 30/100	118.6%	119.0%	1.954	-20.0%
Level 1	Final	SMA 30/100	-7.4%	-5.4%	-0.171	-18.5%
Level 2	Validation	Ensemble	2.8%	119.0%	1.110	-2.3%
Level 2	Final	Ensemble	-0.6%	-5.4%	-0.522	-1.4%
Level 3	Validation	CVaR	126.1%	128.1%	1.696	-34.7%
Level 3	Final	CVaR	-18.0%	-25.4%	-0.023	-45.2%
Level 4	Validation	Calendar	3.4%	63.9%	0.475	-6.6%
Level 4	Final	Calendar	-4.1%	-9.3%	-0.880	-9.1%
Level 5	Validation	Large universe	-18.6%	-7.1%	-4.239	-18.8%
Level 5	Final	Large universe	-28.0%	-45.2%	-0.218	-42.2%

2. Reproducibility, data and benchmark definitions

The accepted lock freezes validation-selected methodology before final-test exposure. This notebook reads committed artifacts; it does not rerun `make final-test` or retune any final-year decision. Public repository visibility must still be verified by the human owner; this local checkout currently has no public remote configured.

Reproducibility and provenance

Metric	Value
Python	3.11
Package manager	uv with committed uv.lock
Final-test period	2025-01-01 to 2025-12-31
Train/validation	2021-2023 train, 2024 validation
Final-test lock	c33b5eb396f6...43af5e
Data SHA-256	9f539f383942...0d7e14
Validation config	da1dcaf44251...2e77e8
Locked commit	d200df6d8a5b...43a1e5
Runner commit	d200df6d8a5b...43a1e5
Base tag	stage/09-level-5-100pairs
Public URL	not configured in this checkout; owner must publish/verify

Data card

Metric	Value
Source	Binance spot USDT via CCXT snapshot
Timeframe	1d OHLCV, UTC bar-start semantics
Period	2021-01-01 to 2025-12-31
Rows	158,511
Symbols	163
Duplicate symbol-days	0
Null OHLCV fields	0
Known limitation	active-market survivorship/delisting bias

What the notebook recomputes versus loads

Section	Behavior
Data validation	read-only verification from committed data/artifacts
Model specification	read from source config and frozen lock
2024 validation	loaded as pre-exposure validation artifacts
2025 final results	loaded as frozen final-test evidence
Final test	never rerun by this notebook

Lock lineage and source of truth

Field	Value
Current release source of truth	c33b5eb396f6...43af5e
Lock role	pretest/final methodology lock for this committed release
Pre-exposure statement	results_inspected=False; state=LOCKED
Locked commit	d200df6d8a5b...43a1e5
Runner commit	d200df6d8a5b...43a1e5
Final artifacts reference	all committed final-test metadata in c33b5eb396f6 references this lock
Earlier external audit reference	dab407... is not a committed release lock in this checkout

Reviewer runbook

Step	Command
Setup	uv sync --frozen
Data	make validate-data
Quality	make lint && make test
Notebook	make notebook-full
Deck	make presentation
Release	make release-verify
Do not rerun	make final-test after exposure

Benchmark definitions

Scope	Benchmark definition
Level 3	broker_costed_equal_weight_static_basket
Level 4	broker_costed_level3_static_benchmark
Level 5	broker_costed_equal_weight_top_k_universe
Levels 1-2	broker_costed_buy_and_hold

Level 5 benchmark reconciliation

Benchmark	Return	Status	Definition
Level 5 frozen final	-45.2%	locked in c33 final-test lock	broker_costed_equal_weight_top_k_universe
BTC buy-and-hold final	-5.4%	diagnostic comparator only	broker_costed_buy_and_hold
Level 5 validation feasibility	-7.1%	short Dec-2024 validation diagnostic	price_normalized_btc_open_to_open

3. Architecture and causal clock

Shared pipeline: frozen OHLCV -> validation gate -> causal features -> typed agents -> aggregator -> pre-risk -> allocator -> rebalance controller -> post-risk -> orders/fills -> ledger -> metrics and monitoring. Features use completed daily bars; execution is at the next available open.

Representative Level 2 agent trace

Agent	Score	Conf	Fit	Action
sma_crossover	-1.000	0.228	2024-12-31	ok
econometric_ar_garch	0.037	0.049	2024-12-31	ok
ml_logistic	-0.128	0.128	2024-12-31	ok
ml_hist_gradient_boosting	-0.236	0.236	2024-12-31	ok
aggregator	-0.466	0.040	2024-12-31	ok
post_risk	approve	cash=100.0%	2025-01-01	ok

4. Level 1 — Baseline Strategy for a Single Cryptocurrency.

Level 1 is a one-symbol configuration of the same broker, risk and ledger engine used by the later levels. The baseline is intentionally simple, then wrapped into the agent interface used by Level 2.

SMA baseline and agent evolution

Metric	Value
Rule	BTC/USDT SMA fast=30, slow=100
Execution	completed close signal; next-open broker execution
Agent evolution	SMA rule -> typed SignalAgent -> score/confidence -> ensemble input
Risk layer	pre/post risk controls decide final exposure and can move to cash

Level 1 validation

Net	Sharpe	MDD	Turn	Cost	Bench
118.6%	1.954	-20.0%	500.5%	$10,771	119.0%

Level 1 final

Net	Sharpe	MDD	Turn	Cost	Bench
-7.4%	-0.171	-18.5%	598.7%	$8,906	-5.4%

5. Level 2 — Adding AI Agents, Econometrics and ML.

Level 2 adds technical, econometric and ML agents on BTC/USDT. The ensemble was frozen as the assignment-representative multi-agent candidate with acceptable validation risk and turnover; it was not selected as the unconditional maximum-performance row.

Model specification

Model	Preprocessing	Parameters	Refit
SMA	completed close history	10/50 in Level 2; 30/100 in Level 1	rule-based
AutoReg/GARCH	open returns	AutoReg + GARCH(1,1)	daily_causal
Logistic Regression	median imputer + StandardScaler	max_iter=500; class_weight=balanced	monthly expanding
HistGradientBoosting	median imputer	60 iter; LR 0.05; 15 leaves; L2 0.01	monthly expanding

Target, features and retraining cadence

Metric	Value
Target	1[open(t+2) / open(t+1) - 1 > 0.0020]
Threshold formula	(10 fee bps + 5 slippage bps + 5 safety bps) / 10000
Feature groups	returns, momentum, trend, RSI/MACD, ATR/range, vol, drawdown, volume
Causal clock	completed daily bars; decisions execute at next open
ML retraining	monthly
Econometric refit	daily_causal
Agent weights	{"econometric": 0.25, "hist_gradient_boosting": 0.25, "logistic": 0.25, "technical": 0.25}

No-leakage fit audit

Metric	Value
Fit-audit rows	2,548
Prediction rows	1,092
Future-label flags	0
Audit status	ok
ML refit	daily_causal, monthly
Feature cutoff example	2025-01-01
Fit cutoff example	2024-12-31

6. Level 2 - validation, predictive metrics and robustness

Validation and final evidence are shown separately to make the selection discipline explicit. Robustness rows distinguish pre-freeze validation evidence from post-hoc final-test diagnostics. Predictive metrics demonstrate the main research result: validation signal did not generalize reliably in the exposed final year.

Level 2 validation candidates

Approach	Selected	Net	Sharpe	MDD	Turn	Cost	Bench
SMA	no	45.2%	1.133	-34.3%	1338.5%	$23,529	119.0%
AR/GARCH	no	2.7%	0.588	-3.4%	2418.0%	$37,110	119.0%
Logistic	no	1.4%	1.415	-0.3%	237.2%	$3,591	119.0%
HGB	no	2.8%	1.353	-1.0%	576.7%	$8,800	119.0%
Ensemble	yes	2.8%	1.110	-2.3%	263.7%	$4,019	119.0%

Level 2 final candidates

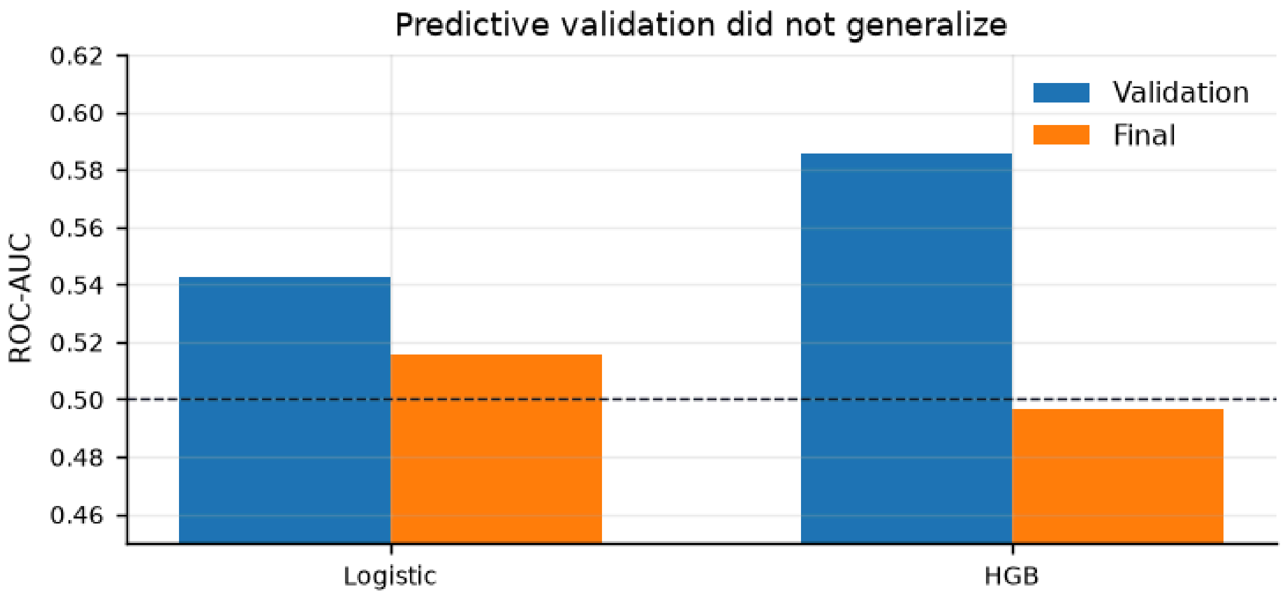
Approach	Selected	Net	Sharpe	MDD	Turn	Cost	Bench
SMA	no	-7.9%	-0.383	-18.9%	915.6%	$13,915	-5.4%
AR/GARCH	no	-3.7%	-1.411	-4.1%	2033.8%	$29,711	-5.4%
Logistic	no	-0.4%	-0.602	-0.6%	193.6%	$2,900	-5.4%
HGB	no	0.0%	0.020	-1.3%	462.3%	$6,947	-5.4%
Ensemble	yes	-0.6%	-0.522	-1.4%	106.5%	$1,600	-5.4%

Predictive metrics

Model	Split	ROC-AUC	PR-AUC	Log loss	Brier
Logistic	Validation	0.542	0.545	0.689	0.248
HGB	Validation	0.586	0.558	0.690	0.248
Logistic	Final	0.515	0.501	0.689	0.248
HGB	Final	0.497	0.453	0.706	0.256

Robustness and randomness checks

Split	Role	Period	Sharpe 95% CI	Shift p	Corr	Net ROI
Validation	used before freeze	2024-01-01 to 2024-12-31	-1.05 to 3.02	0.682	-0.0177	2.8%
Final	post-hoc diagnostic only	2025-01-01 to 2025-12-31	-2.30 to 1.42	0.984	-0.0021	-0.6%



7. Level 3 — Portfolio Management on Historical Data.

Level 3 uses a frozen 5-7 asset universe and an exact trailing 12-month final estimation window. Candidate methods are compared on validation; the final 2025 run executes the frozen method without switching to a better-looking final comparator.

Level 3 construction and trading path

Metric	Value
Final assets	BTC/USDT, ETH/USDT, SOL/USDT, PEPE/USDT, BNB/USDT, XRP/USDT, DOGE/USDT
Estimation window	2024-01-01 to 2024-12-31
Candidate methods	equal_weight, inverse_volatility, minimum_variance, cvar_downside
Selected method	cvar_downside
Objective	validation net Sharpe; risk/cost metrics reported net of fees/slippage
Trading application	target weights -> risk approval -> next-open orders/fills -> ledger

Selected CVaR weights

Symbol	Weight
BTC/USDT	19.8%
BNB/USDT	18.2%
ETH/USDT	15.7%
XRP/USDT	13.4%
SOL/USDT	12.9%
DOGE/USDT	10.8%
PEPE/USDT	8.7%
Cash/rounding	0.5%

Level 3 validation candidates

Method	Selected	Net	Sharpe	MDD	Turn	Cost	Bench
Equal weight	no	128.1%	1.676	-36.3%	99.5%	$1,492	128.1%
Inv vol	no	121.8%	1.667	-35.2%	99.5%	$1,492	128.1%
Min var	no	111.4%	1.653	-32.8%	99.5%	$1,492	128.1%
CVaR	yes	126.1%	1.696	-34.7%	99.5%	$1,492	128.1%

Level 3 final candidates

Method	Selected	Net	Sharpe	MDD	Turn	Cost	Bench
Equal weight	no	-25.4%	-0.125	-48.7%	99.5%	$1,492	-25.4%
Inv vol	no	-16.4%	0.002	-44.8%	99.5%	$1,493	-25.4%
Min var	no	0.9%	0.280	-36.3%	99.5%	$1,493	-25.4%
CVaR	yes	-18.0%	-0.023	-45.2%	99.5%	$1,493	-25.4%

8. Level 4 — Dynamic Portfolio Rebalancing.

Calendar monthly was the frozen selected policy. Drift and signal/risk policies can show better final return, but the post-exposure rule is strict: do not replace the selected policy after seeing final-year results.

Level 4 policy mechanics

Policy	Calendar	Drift	Score trigger	Risk trigger
Calendar	monthly	disabled	disabled	no
Drift	monthly	5.0%	disabled	no
Signal/risk	monthly	5.0%	15.0%	yes

Level 4 validation candidates

Policy	Selected	Net	Sharpe	MDD	Turn	Cost	Bench
Static	no	126.1%	1.696	-34.7%	99.5%	$1,492	126.1%
Calendar	yes	3.4%	0.475	-6.6%	182.0%	$2,763	63.9%
Drift	no	55.1%	1.513	-22.8%	931.7%	$13,040	63.9%
Signal/risk	no	55.1%	1.513	-22.8%	931.7%	$13,040	63.9%

Level 4 final candidates

Policy	Selected	Net	Sharpe	MDD	Turn	Cost	Bench
Static	no	-18.0%	-0.023	-45.2%	99.5%	$1,493	-18.0%
Calendar	yes	-4.1%	-0.880	-9.1%	237.3%	$3,584	-9.3%
Drift	no	3.8%	0.290	-23.5%	839.0%	$12,733	-9.3%
Signal/risk	no	3.8%	0.290	-23.5%	839.0%	$12,733	-9.3%

9. Level 5 — Portfolio Expansion to 100+ Pairs.

Level 5 is a deterministic cross-sectional quant scoring agent, not a fitted ML model. It proves the 100+ pair requirement and records operational health beyond trading KPIs. Its validation window is a short late-December 2024 mechanical and scalability proof, not strong statistical evidence of investment quality.

Level 5 mechanics

Question	Implementation
Pair selection	USDT spot; history/liquidity filters; excludes stable/fiat bases
Signal priority	deterministic cross-sectional factor score
Factors	20/60-day momentum, vol, drawdown, liquidity penalties
Portfolio	top 25; max 5%; inverse_volatility allocation
Rebalance	weekly plus drift/signal/risk; score trigger 10%
Risk	pre/post gates, cash approval, caps, turnover checks
Monitoring	runtime, RSS, incidents, fallback, nulls, pair counts
Fail-safe	abstain, retain weights, move to cash, kill switch evidence
Long-term quality	freshness, coverage, feature drift, calibration, disagreement, rolling Sharpe/drawdown, cost drift, universe churn

Level 5 validation summary

Net	Sharpe	MDD	Turn	Cost	Bench
-18.6%	-4.239	-18.8%	346.3%	$5,216	-7.1%

Level 5 final summary

Net	Sharpe	MDD	Turn	Cost	Bench
-28.0%	-0.218	-42.2%	4924.5%	$110,939	-45.2%

Monitoring and incidents

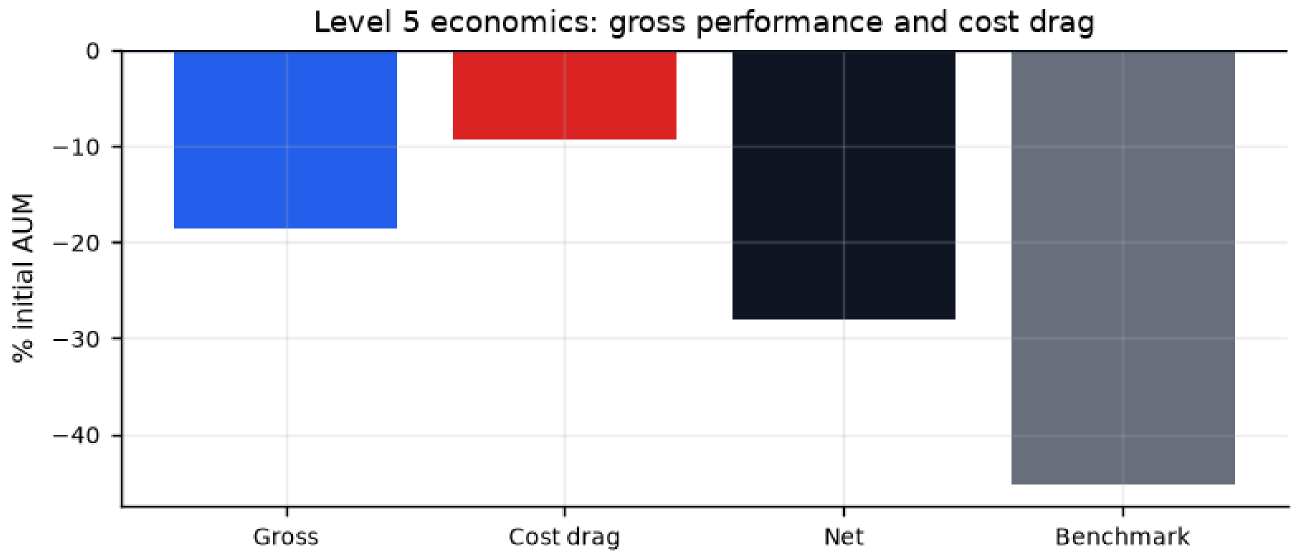
Metric	Value
System status	ok
Status meaning	operational pipeline health, not investment performance
Recorded monitoring events	53
Critical alerts	0
Warning alert rows	2
Info alert rows	2
Invalid data count	0
Invalid model count	0
Optimizer fallback rate	14.5% final-test rate
Abstention rate	0.0%
Fail-safe evidence	kill_switch_cash_schedule_unit_test, volatility_limit_cash_approval, invalid_feature_alert_path, weight_reconciliation_post_risk

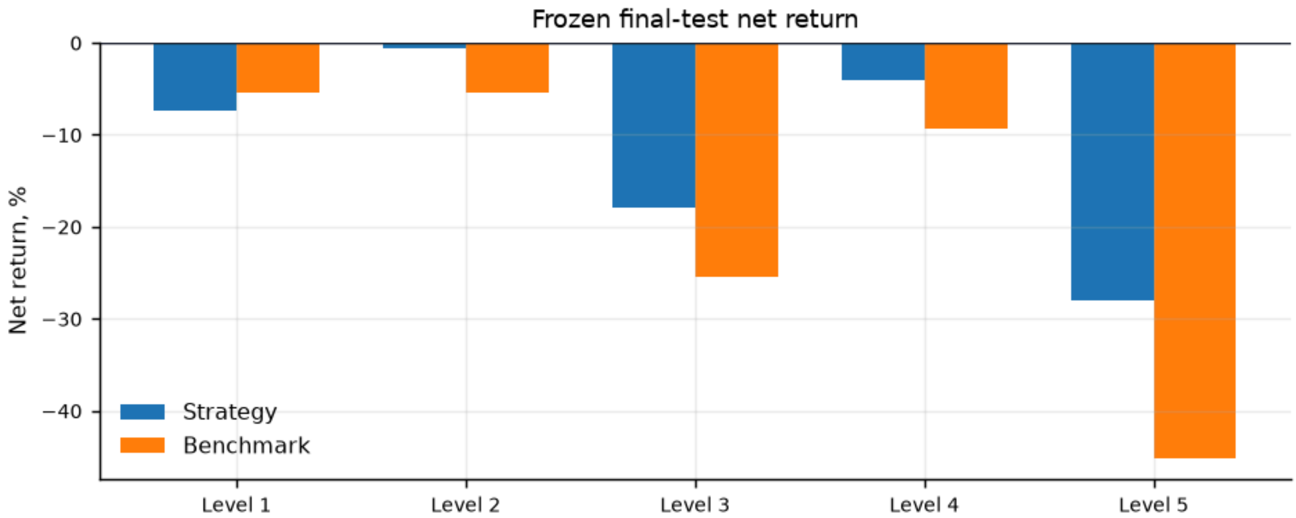
10. Level 5 economics and cross-level diagnostics

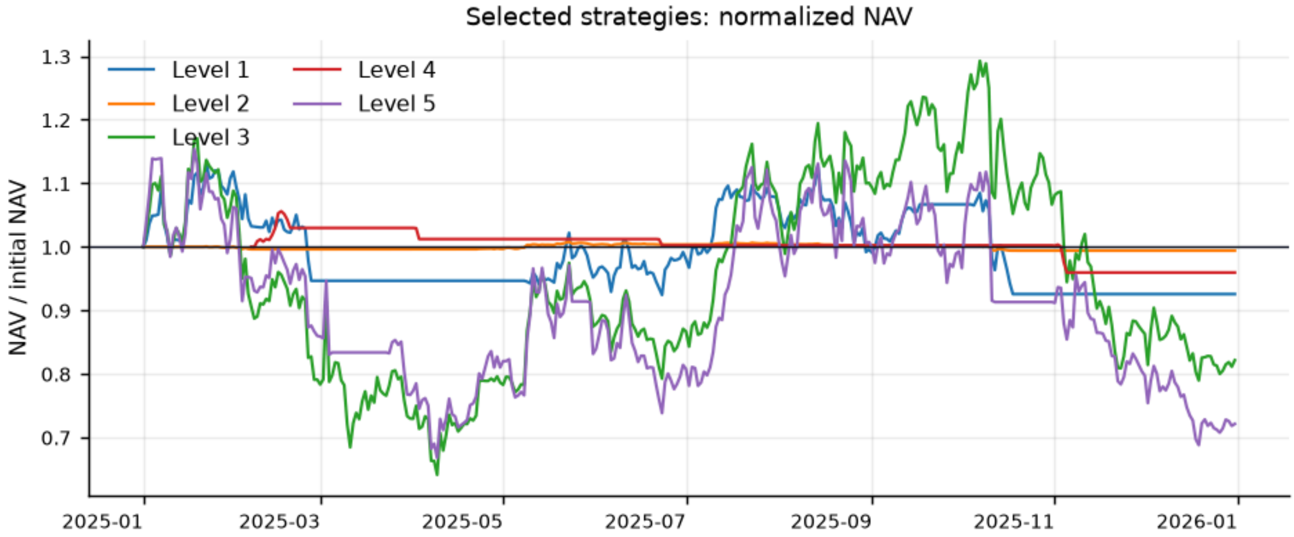
Scalable processing was achieved, but the factor policy produced negative gross performance and very high turnover. Costs materially worsened the result. This is a central research conclusion, not a hidden failure.

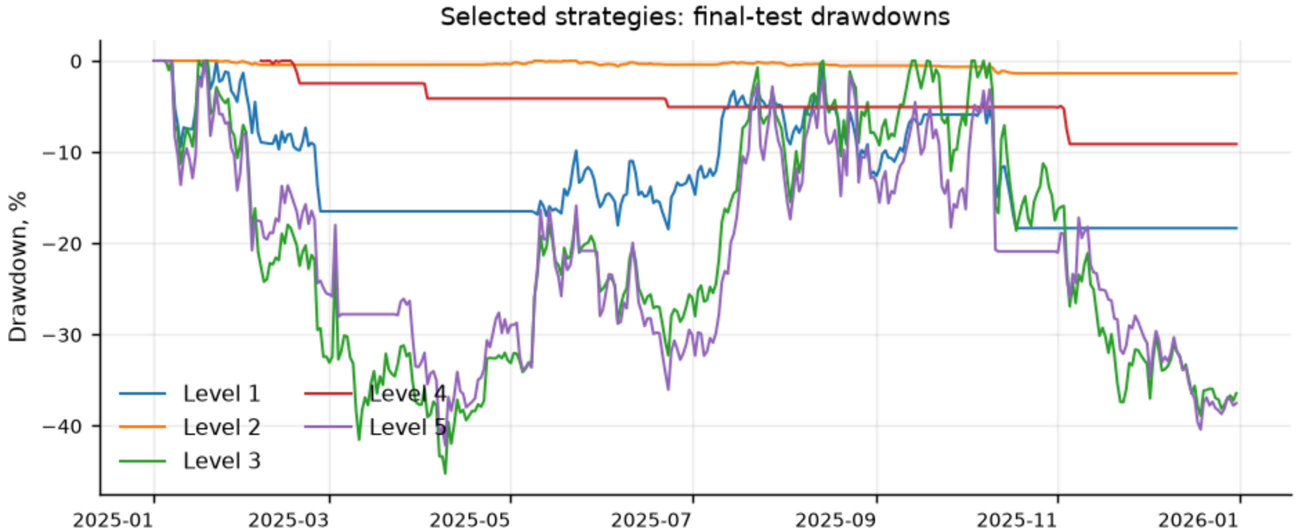
Level 5 cost and turnover

Metric	Value
Gross return	-18.6%
Return drag from costs	9.4% gross-to-net
Net return	-28.0%
Benchmark	-45.2%
Rebalance decisions	364
Fills	8,727
Fee-bearing notional	$73,959,230
Total cost	$110,939
Cost as % initial AUM	11.1%
Average cash	14.7%









11. Selection rationale, limitations and roadmap

The final suite is exposed, so the only allowed changes are presentation, explanation and visualization of already-frozen artifacts. Limitations: active-market survivorship/delisting bias, daily-bar liquidity proxies, USDT cash assumption, simplified fills, cash-heavy risk behavior, short Level 5 validation window and Level 5 top-K benchmark rather than a full eligible-universe benchmark. Future work includes multi-CEX adapters, order-book liquidity, reconciliation, Telegram controls and news/sentiment ingestion; those future items are not enabled in this MVP.

Validation-selected methodology

Level	Frozen selection	Rationale
1	SMA 30/100 BTC	validation net Sharpe among SMA grid; typed SignalAgent wrapper
2	Ensemble	multi-agent candidate; acceptable risk, not max-performance row
3	cvar_downside	maximum validation net Sharpe in static portfolio candidates
4	calendar_monthly	validation selected under MDD <= 25%, turnover <= 6.0
5	top-25 inv-vol universe	cross-sectional scoring; validates >=100 pairs and fail-safes